

Computer Vision for Robotics

Computer Vision

Computer vision allows robots to extract useful information from images and video for perception and decision-making.

Image Processing

Common operations include resizing, filtering, color conversion, thresholding, edge detection, and feature extraction.

Object Detection

Object detection identifies objects and estimates their locations, often using bounding boxes.

Object Tracking

Object tracking estimates the movement of detected objects across consecutive video frames.

Applications

Computer vision is used in autonomous navigation, inspection, manipulation, surveillance, agriculture, and human-robot interaction.